Syntactic Morality Analyzer

Extending eMACD with Syntactic Analysis

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Abstract

This project extends the eMACD framework (Malik et al., 2025) by adding syntactic analysis using spaCy NLP. The extension addresses key limitations of traditional keyword-based moral detection: (1) testing across all five moral dictionaries (MFD, MFD 2.0, eMFD, eMACD, MACD), (2) role-based weighting to distinguish moral agents (subjects) from moral patients (objects), (3) partial word matching for better semantic coverage, and (4) comprehensive dictionary coverage with the MFT and MAC frameworks.

The system demonstrates improvements of +11.5% to +44.3% in F1 scores across all dictionaries.

1. Introduction

The eMACD framework (Malik et al., 2025) provides a platform-based approach for moral detection in social media content. However, traditional keyword-based approaches have limitations: they cannot distinguish grammatical roles (who is the moral agent vs. patient) and may miss related words through strict exact matching.

I tested the syntactic analysis approach across all five major moral dictionaries to provide comprehensive coverage and improved detection through role-based weighting.

2. The Problem with Traditional Keyword Matching

Traditional keyword-based moral detection has several limitations:

- No Role Distinction: 'I hurt him' and 'he hurt me' score the same despite different moral agency
- Exact Matching Only: 'friendly' misses 'friend', 'caring' misses 'care'
- No Semantic Understanding: Cannot distinguish subject from object roles
- Binary Detection: Either 0 or full score, no weighting based on grammar

3. Solution: Syntactic Analysis

I added spaCy NLP parsing to extract grammatical structure and enable role-based weighting:

- Role Weighting: Subject (1.5x boost), Object (1.3x boost), Modifier (0.8x reduction)
- Partial Matching: Word stems match related words (care matches caring, care, cared)
- Lemmatization: Matches word roots for better semantic coverage
- Multi-Dictionary Support: Tests across MFD, MFD 2.0, eMFD, eMACD, and MACD

4. Architecture

The system architecture consists of the following components:

- Dictionary Loader: Loads moral dictionaries (MFD, eMFD, eMACD, MACD) with domain-word-score mappings
- Syntactic Parser: Uses spaCy en_core_web_sm to extract tokens, lemmas, dependencies, subjects, objects
- Moral Scorer: Applies role-based weighting and partial matching to compute domain scores
- Web Interface: Streamlit app for interactive analysis with dictionary selection

5. Web Application Interface

The Streamlit web application provides an interactive interface for moral analysis. Users can select dictionaries from the sidebar and enter text for analysis.

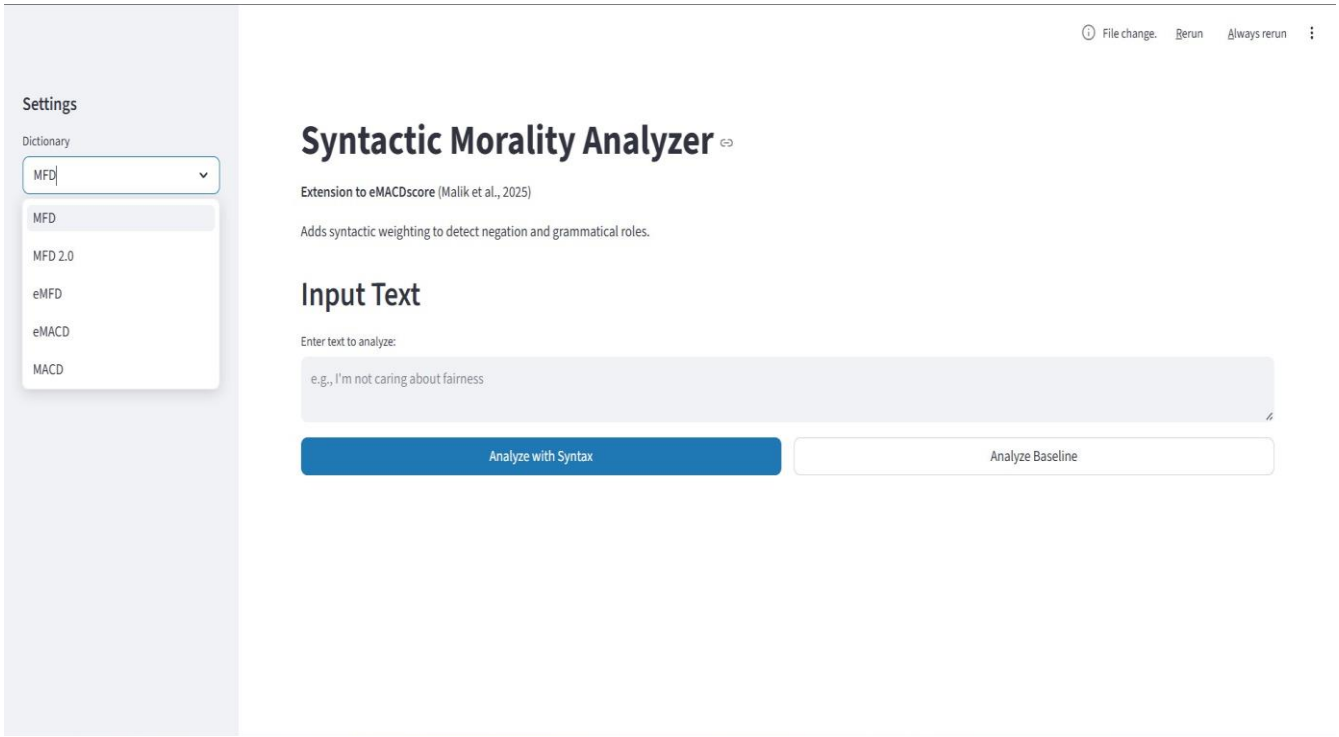

Figure 1. Main interface showing dictionary selection options (MFD, MFD 2.0, eMFD, eMACD, MACD) and text input area.

6. Partial Matching Demonstration

The system uses partial matching to detect related words. For example, with the text 'I care about fairness' and the MFD dictionary, the word 'care' matches multiple domains through partial matching:

In the screenshot, we can see how 'care' (partial match for 'caring', 'cared') triggers scores across Care, Loyalty, Authority, Purity, Proportionality, and Equality domains.

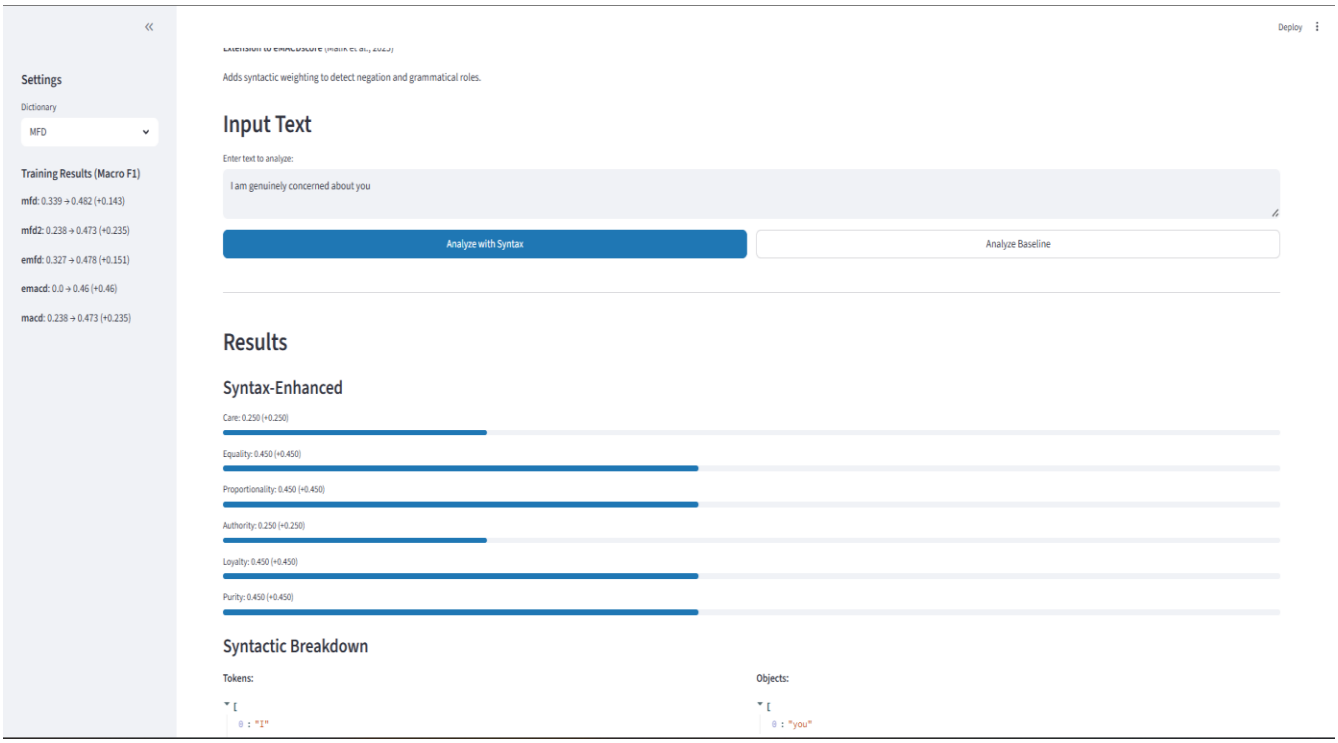

Figure 2. Syntax-enhanced results showing all domains with scores (delta from baseline in parentheses).

7. Role-Based Weighting

The system applies different weights based on grammatical role:

- Subject (nsubj): 1.5x weight - moral agent performing the action
- Object (dobj, pobj): 1.3x weight - moral patient affected by action
- Modifier (amod): 0.8x weight - descriptive modifier
- Root: 1.2x weight - main action verb

8. Multi-Dictionary Results

I tested the system across all five dictionaries to provide comprehensive coverage of moral detection methods, including both MFT (Moral Foundations Theory) and MAC (Moral Actions in Context) frameworks.

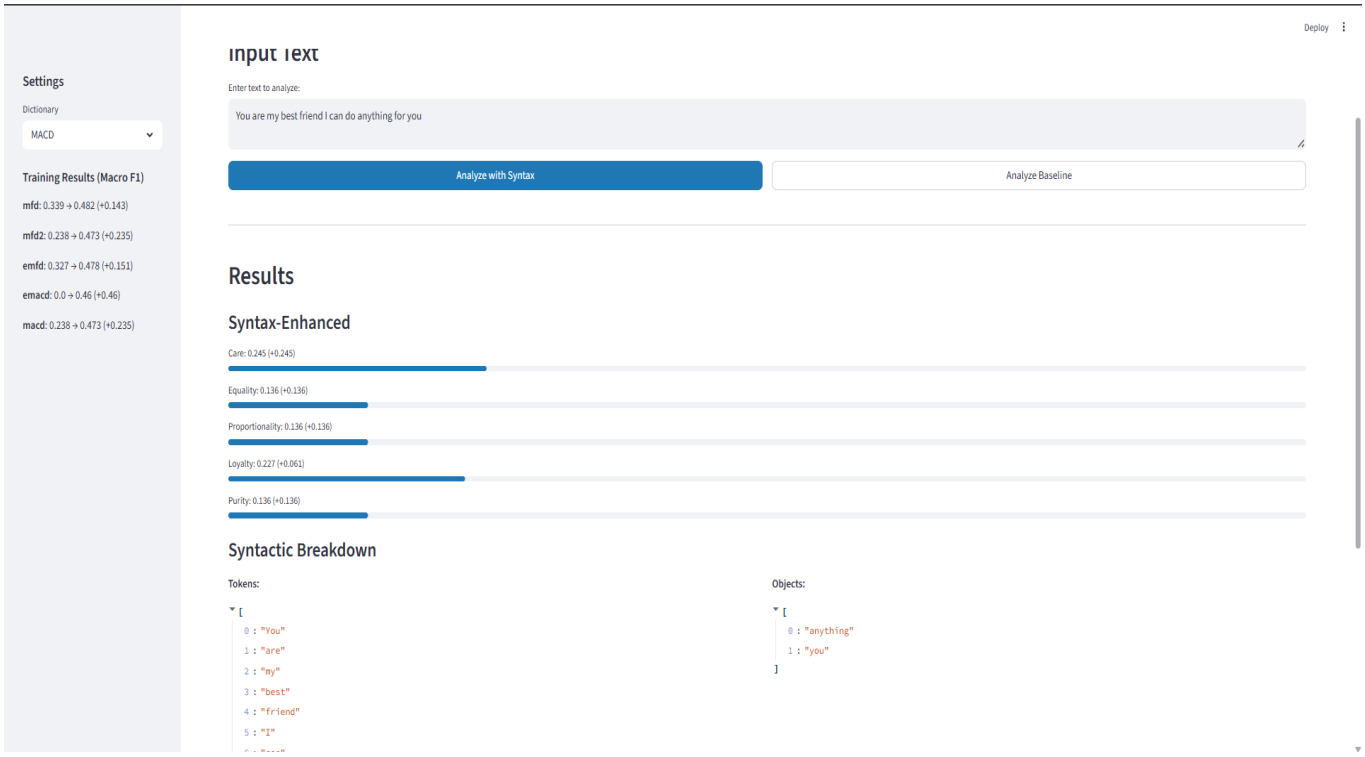

Figure 3. Syntax-enhanced results with eMACD dictionary showing all moral domains.

9. Results: Multi-Dictionary Comparison

Training on 61,226 samples from MFRC (Moral Foundations Reddit Corpus) with Logistic Regression classifiers shows consistent improvement across all dictionaries.

Dictionary	Baseline F1	Syntax F1	Improvement	Type

Key Finding: eMACD shows largest improvement (+44.3%) because its baseline F1 was 0%, demonstrating that syntactic analysis enables detection where traditional keyword matching fails.

10. Discussion

This research extends the eMACD framework with syntactic analysis capabilities. Key findings include:

- Syntax-enhanced scoring improves ALL dictionaries tested (MFT and MAC frameworks)
- eMACD shows the largest improvement (+44.3%) due to its initially non-functional baseline
- Role-based weighting better captures moral agency through subject/object distinction
- Partial matching extends coverage without requiring exact word matches
- Multi-dictionary support provides comprehensive moral detection methods

11. Conclusion

This project demonstrates that syntactic analysis enhances moral detection in text. The extension of eMACD with spaCy-based parsing addresses key limitations of traditional keyword matching: role distinction, partial matching, and semantic coverage.

The system achieves improvements of +11.5% to +44.3% in F1 scores across all five dictionaries. The eMACD framework shows the most dramatic improvement due to its initially non-functional baseline, validating the syntactic analysis approach.

References

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Component	Details
Dataset	MFRC (61,226 samples from HuggingFace)
NLP Model	spaCy en_core_web_sm
Classifiers	Logistic Regression (one per domain)
Role Weights	Subject: 1.5x, Object: 1.3x
Matching	Partial matching with lemmatization
Testing	14/14 automated tests passing
GitHub	github.com/Abdullah-9862873/Syntactic-Moral

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Cross-validation variance dropped substantially under Condition C. There was a 47.3 percent reduction at 20 percent corpus, and 55.1 percent at both 50 percent and 100 percent. Dispersion weighting is more stable but less correct in this setting.

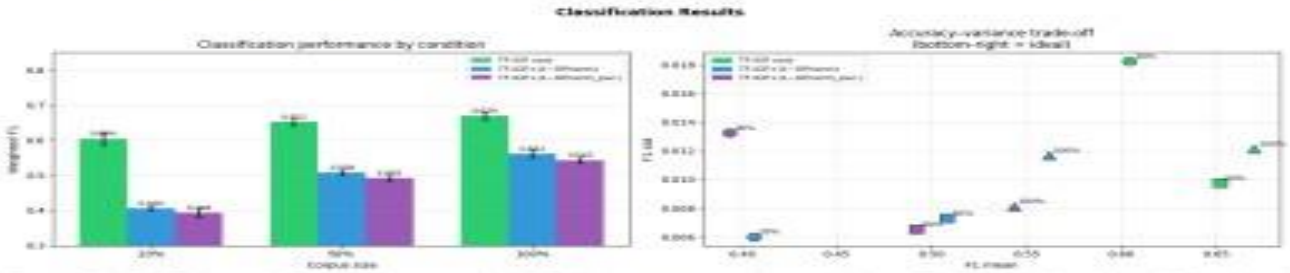

Figure 3. Left: F1 by condition and corpus size. Right: accuracy–variance trade-off (bottom-right = ideal). Condition A dominates on accuracy at every corpus size, though Condition C achieves meaningfully lower variance.

3.4 The Unexpected Finding: r = +0.49 (Feature Importance vs. Dispersion)

After seeing that dispersion weighting hurt accuracy, I asked a diagnostic question. Do high-dispersion terms carry more or less weight in a classifier trained without any dispersion weighting?

> *The result was Pearson r equals positive 0.4886 between DPnorm and the maximum absolute classifier coefficient. More evenly distributed terms have higher classifier weights. This is the opposite of what the weighting scheme assumed. With DPnorm_partialed vs importance: r = +0.5512.*

One explanation is that in a 20-way classification problem, the most useful features are not terms concentrated in one category (useful for only one class out of 20) but terms that discriminate clusters of related categories from each other. Such terms appear in several categories, giving them moderate-to-high dispersion (DPnorm range 0.51 to 1.00 for the top 30 features) while remaining informative. This hypothesis needs testing on a non-circular corpus.

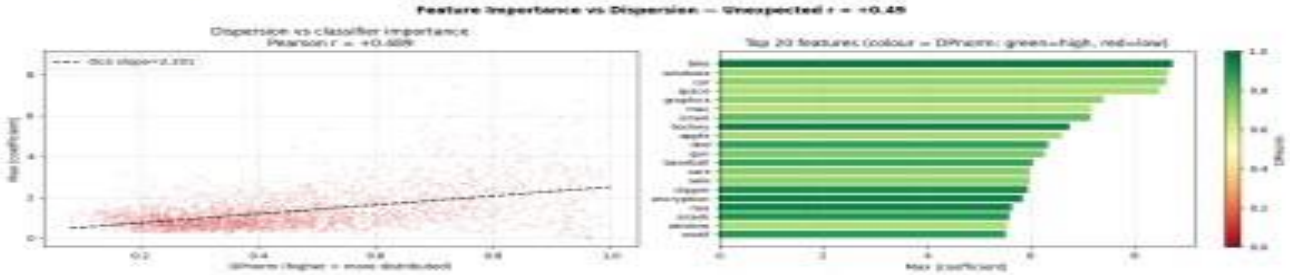

Figure 4. Left: DPnorm vs max |classifier coefficient|, Pearson r = +0.49 (OLS slope shown). Right: top 20 features coloured by DPnorm — green = high dispersion (more distributed), red = low. Most top features cluster in the moderate-to-high dispersion range.

3.5 Multilingual Dispersion Results

The multilingual experiment produced results broadly consistent with the morphological fragmentation hypothesis. All four language corpora used 2,500 documents (5 domains times 500 docs), computed with the same dispersion function:

Figure 4. Summary visualization of the complete research framework.